1. Introduction

This report contains the results obtained in the emotion prediction task for different datasets. The results are shown based on the best models obtained for each stage and textual representation.

- **Databases:** (1) TweetEmotions[1]; (2) PlayStoreReviews[2]; (3) GoEmotions[3].
- **Training/Validation/Testing Ratio**: 70/15/15.
- **Text Representations**: BoW, TF-IDF (TFIDF).
- **Classification Models**: Logistic Regression, Linear SVC, Random Forest Classifier[4], XGB Classifier, and BERTimbau-based.
- **Evaluation Metrics**: Accuracy, Precision, Recall, and F1, in macro, weighted, and micro parameters.
- **Cross-validation**: 5 folds * 6 times.
- **Observation**: the seed used was 42 in all cases.

1.1 Text Preprocessing

1. Handling missing values.
2. Text normalization and cleaning (BASE_TEXT):
   a. Removal of duplicate spaces and line breaks;
   b. Removal of HTML, URLs, and email addresses;
   c. Accent normalization;
   d. Removal of technical alphanumeric tokens;
   e. Removal of isolated punctuation;
   f. Conversion to lowercase and renormalization of spaces;
3. Linguistic processing with spaCy (CLEAN_TEXT):
   a. Lemmatization (TEXT_LEMMA);
   b. Removal of stopwords (TEXT_NO_STOP).

[1] https://sol.sbc.org.br/index.php/stil/article/view/31160
[2] https://sol.sbc.org.br/index.php/dsw/article/view/30616
[3] https://www.kaggle.com/datasets/antoniomenezes/go-emotions-ptbr
[4] Except for balancing in GoEmotions.

2. Model Hyperparameters

Model	Hyperparameters
LogisticRegression	max_iter=1000, random_state=42
LinearSVC	random_state=42
RandomForestClassifier	random_state=42
XGBClassifier	objective="multi:softprob", eval_metric="mlogloss", random_state=42

BERTimbau-base training hyperparameters

Argument	Value
eval_strategy	“epoch”
save_strategy	“epoch”
learning_rate	2e-5
per_device_train_batch_size	16
per_device_eval_batch_size	16
num_train_epochs	4
weight_decay	0.01
load_best_model_at_end	True
metric_for_best_model	“f1_macro”
logging_steps	100
fp16	True

3. Additional Linguistic Features

3.1 Features Extracted with Textstat

Feature	Observation
letter_count	Simple proxy for intensity/verbosity
sentence_count	In tweets, often = 1; still useful if there is punctuation
avg_letter_per_word	Emotions affect lexical choice
words_per_sentence	Redundant if sentence_count ≈ 1
avg_syllables_per_word	Good proxy for complexity in PT
reading_time	Strongly correlated with letter_count

3.2 Handcrafted features

Feature	Observation
mean_punctuation_per_word	Punctuation can capture emotional information. Exclamation points are associated with high-intensity emotions, such as anger and joy, while question marks generally signal uncertainty, irony, or anxiety
mean_verbs_per_word	Verb density can reveal narrative and dynamism
mean_auxiliaries_per_word	Auxiliary verbs can reflect aspectual and modal constructions
mean_auxiliaries_per_sentence	Redundant if sentence_count ≈ 1
flesch_portuguese	The Flesch PT Readability Index has been widely adopted in readability studies for Brazilian Portuguese and is based on syllabic and sentence-level statistics
uppercase_percentage	Capitalization can function as a paralinguistic marker of emphasis and greater emotional intensity in informal digital communication
mean_unique_word_length	The average length of single words captures lexical sophistication while reducing the influence of repetition
mean_char_repetition_per_word	Repetition of characters within words is a stylistic marker of emotional emphasis, commonly used to intensify affective expression in social media texts

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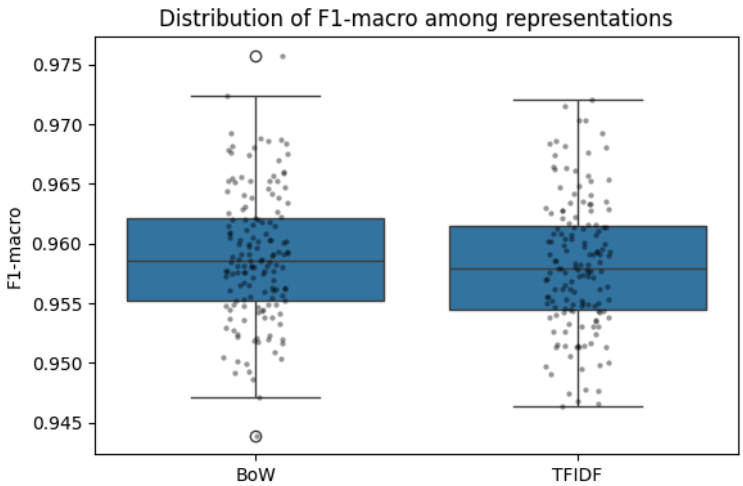
4. TweetEmotions Dataset

I. Baseline (with BASE_TEXT)

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	XGBClassifier	0.962 ± 0.0046	**0.9642 ± 0.0045**	0.9623 ± 0.0045	**0.9628 ± 0.0045**
TFIDF	LinearSVC	0.962 ± 0.0046	0.9629 ± 0.0045	**0.9624 ± 0.0046**	0.9624 ± 0.0046



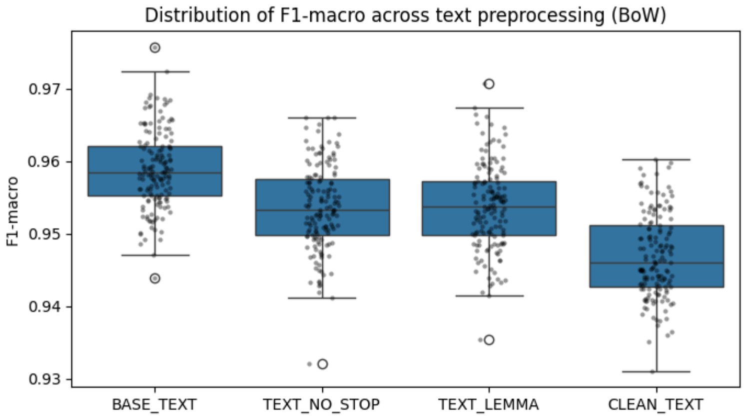
II. Text Preprocessing

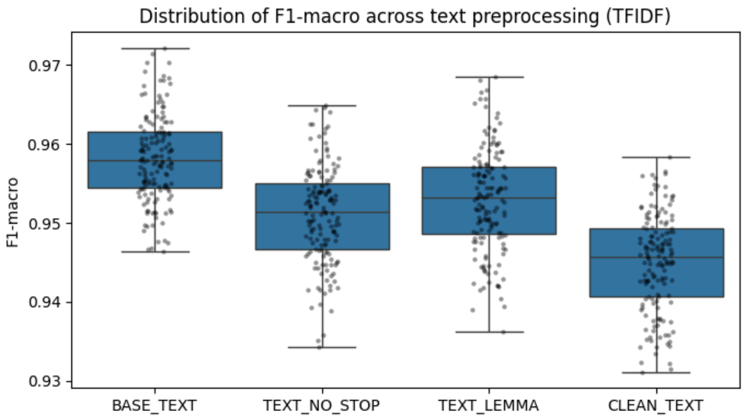
Bow:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	XGBClassifier	**0.962 ± 0.0046**	**0.9642 ± 0.0045**	**0.9623 ± 0.0045**	**0.9628 ± 0.0045**
NO_STOP	XGBClassifier	0.9568 ± 0.0043	0.9631 ± 0.0035	0.9572 ± 0.0043	0958.7 ± 0.004
LEMMA	XGBClassifier	0.9574 ± 0.0043	0.9596 ± 0.0043	0.9578 ± 0.0042	0.9582 ± 0.0042
CLEAN	XGBClassifier	0.9501 ± 0.0048	0.9563 ± 0.0043	0.9505 ± 0.0048	0.9521 ± 0.0045

TFIDF:

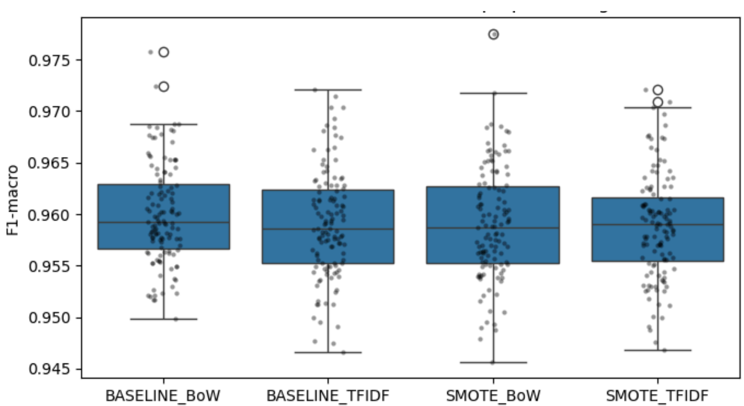
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	LinearSVC	**0.962 ± 0.0046**	**0.9629 ± 0.0045**	**0.9624 ± 0.0046**	**0.9624 ± 0.0046**
NO_STOP	XGBClassifier	0.9524 ± 0.0045	0.9586 ± 0.0038	0.9528 ± 0.0044	0.9543 ± 0.0042
LEMMA	LinearSVC	0.9583 ± 0.0046	0.9593 ± 0.0045	0.9587 ± 0.0046	0.9588 ± 0.0045
CLEAN	XGBClassifier	0.9473 ± 0.0045	0.9532 ± 0.0041	0.9478 ± 0.0045	0.9493 ± 0.0043





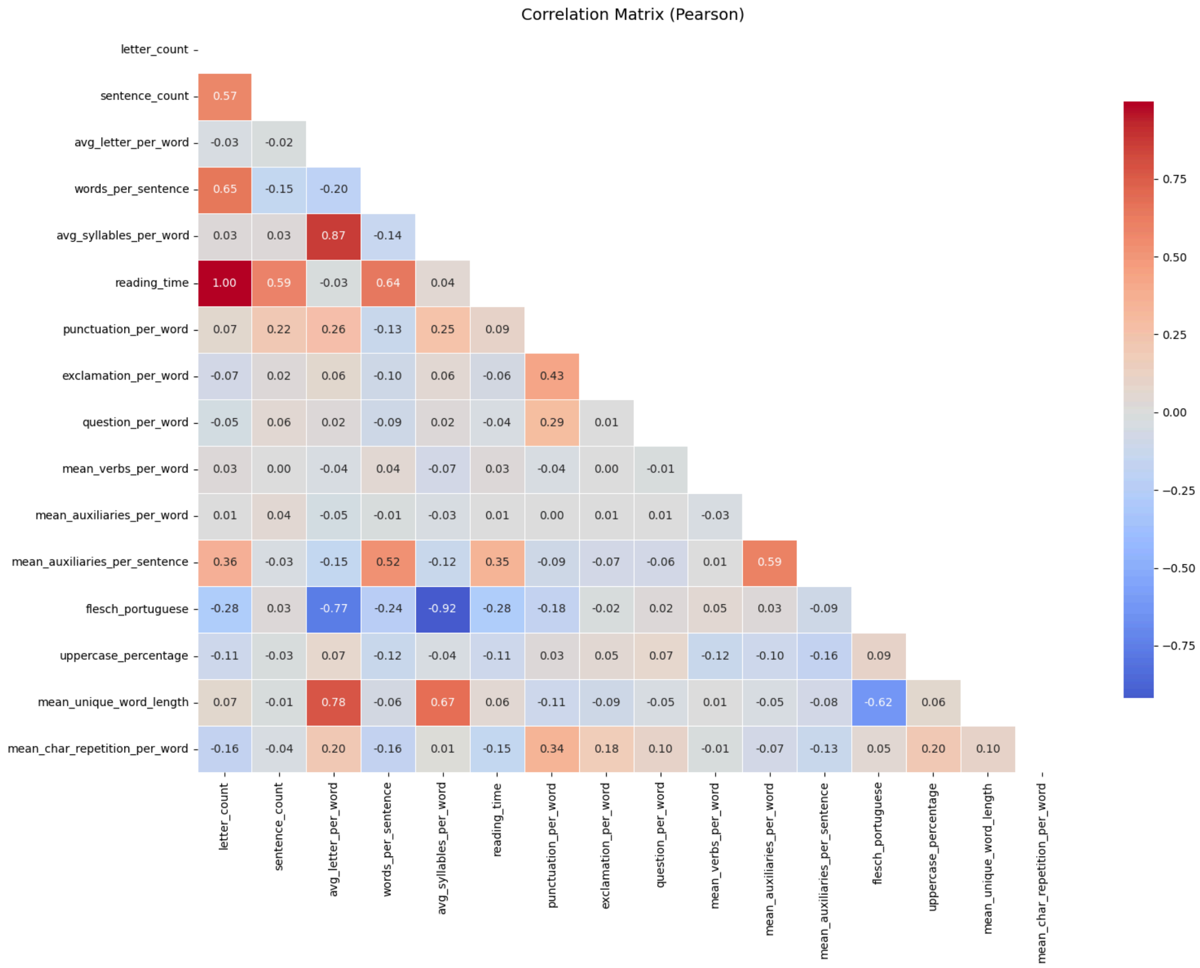
III. Balancing with SMOTE

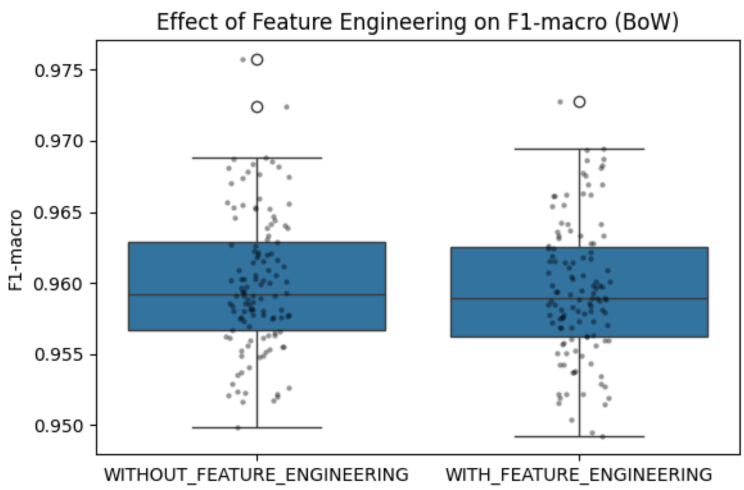
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	XGBClassifier	**0.9623 ± 0.0045**	**0.9645 ± 0.0044**	**0.9626 ± 0.0044**	**0.9631 ± 0.0044**
TFIDF	LinearSVC	0.9619 ± 0.0045	0.9628 ± 0.0044	0.9623 ± 0.0045	0.9623 ± 0.0045

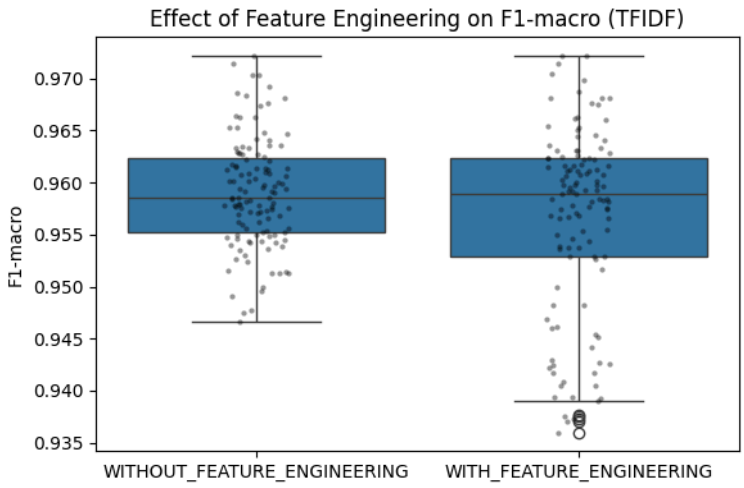


IV. Feature Engineering

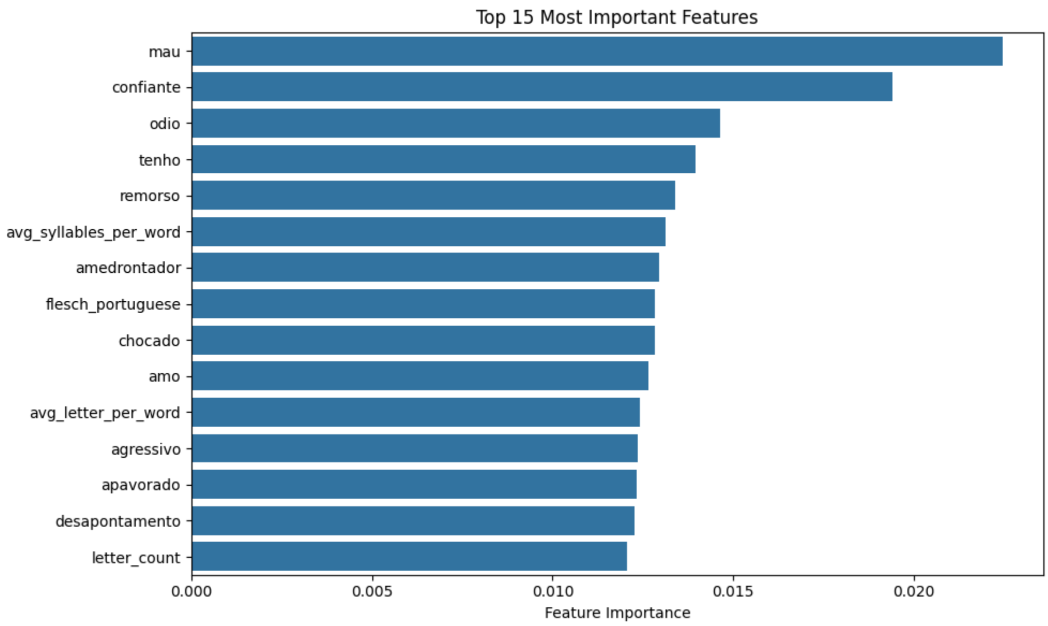
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	XGBClassifier	0.9613 ± 0.0044	**0.9629 ± 0.0043**	0.9616 ± 0.0043	0.9619 ± 0.0043
TFIDF	LinearSVC	**0.9617 ± 0.0047**	0.9628 ± 0.0045	**0.9621 ± 0.0046**	**0.9622 ± 0.0046**

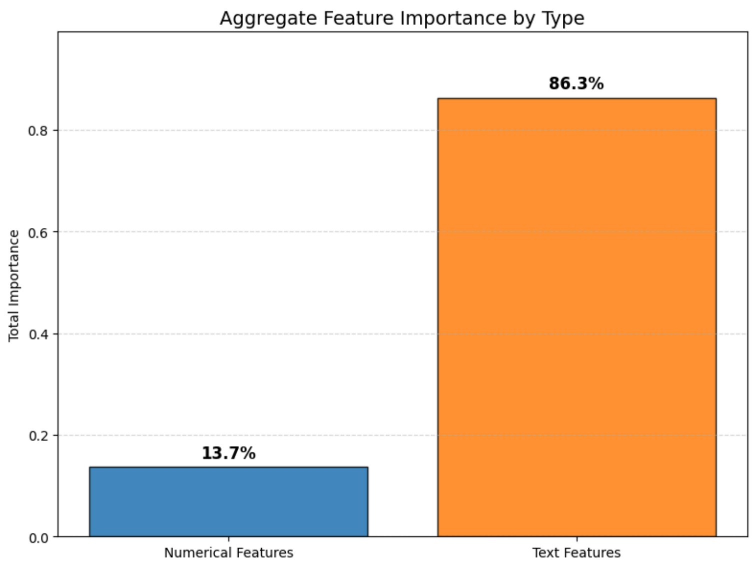




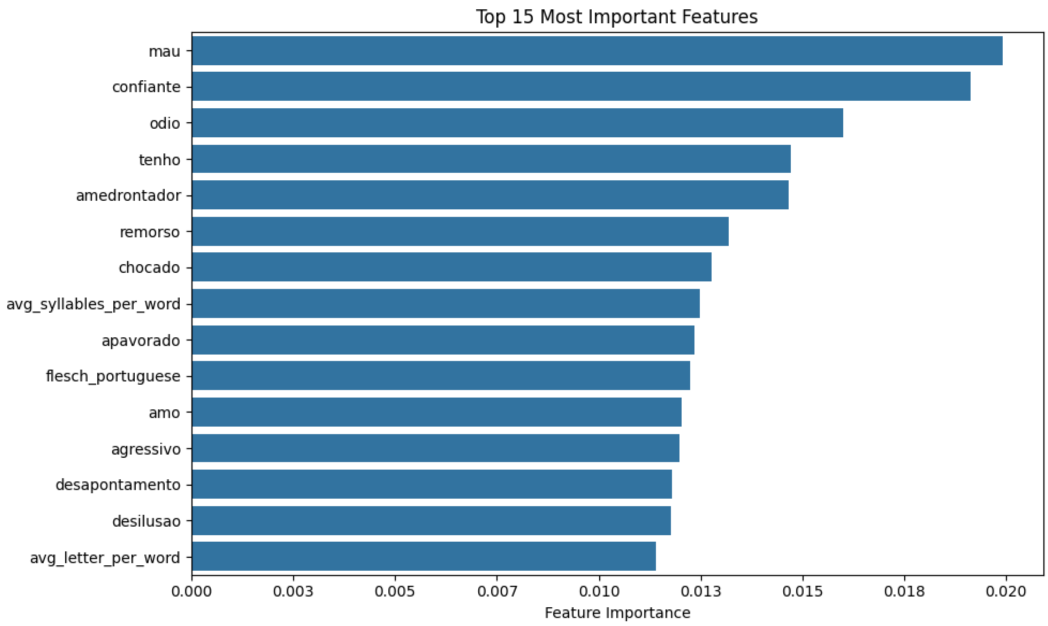

Effect of Feature Engineering on F1-macro (TFIDF)
F1-macro
0.970
0.965
0.960
0.955
0.950
0.945
0.940
0.935
WITHOUT_FEATURE_ENGINEERING
WITH_FEATURE_ENGINEERING

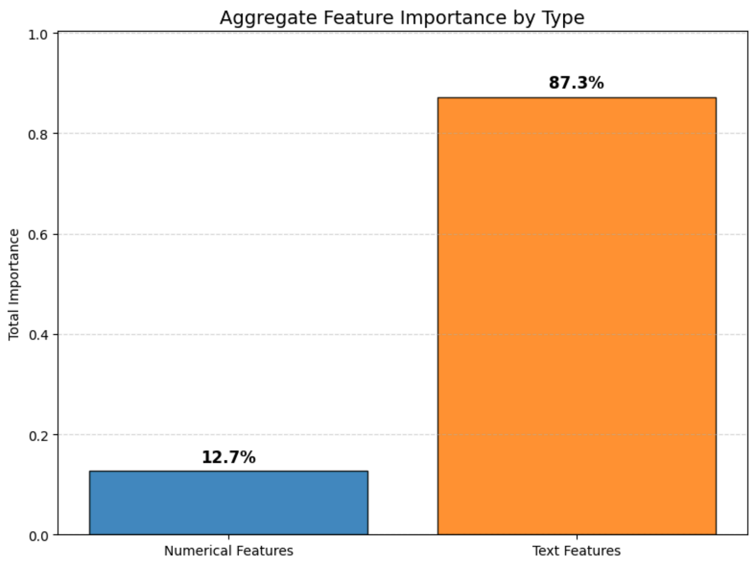
BoW:


Top 15 Most Important Features
mau
confiante
odio
tenho
remorso
avg_syllables_per_word
amedrontador
flesch_portuguese
chocado
amo
avg_letter_per_word
agressivo
apavorado
desapontamento
letter_count
0.000
0.005
0.010
0.015
0.020
Feature Importance


Aggregate Feature Importance by Type
Total Importance
0.0
0.2
0.4
0.6
0.8
13.7%
86.3%
Numerical Features
Text Features

TFIDF:





V. Final Test

Argument	Value
Model	XGBoost
Representation	BoW
Stopwords or Lemmatization?	No
Additional Features?	No
F1 macro (CV)	0.9631 ± 0.0044

Accuracy	Precision_macro	Recall_macro	F1_macro
0.9581	0.959	0.9583	0.9586

VI. BERTimbau Evaluation

Without balancing:

Loss	Accuracy	Precision_macro	Recall_macro	F1_macro
0.1726	**0.9597**	**0.9605**	**0.9602**	**0.9603**

With balancing (CB Loss):

Loss	Accuracy	Precision_macro	Recall_macro	F1_macro
0.1789	0.9565	0.9575	0.9571	0.9572

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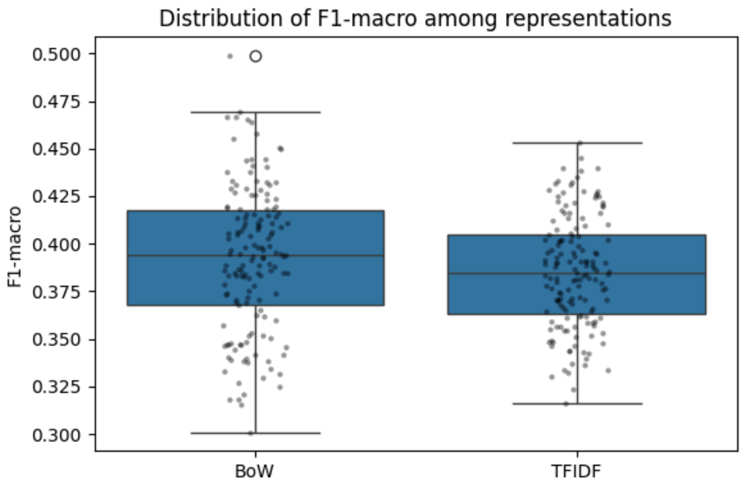
5. PlayStoreReviews

I. Baseline (with BASE_TEXT)

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LogisticRegression	**0.5649 ± 0.021**	**0.4592 ± 0.0498**	**0.4164 ± 0.0249**	**0.4259 ± 0.03**
TFIDF	XGBClassifier	0.5604 ± 0.0216	0.4586 ± 0.0365	0.3976 ± 0.0194	0.4124 ± 0.0224



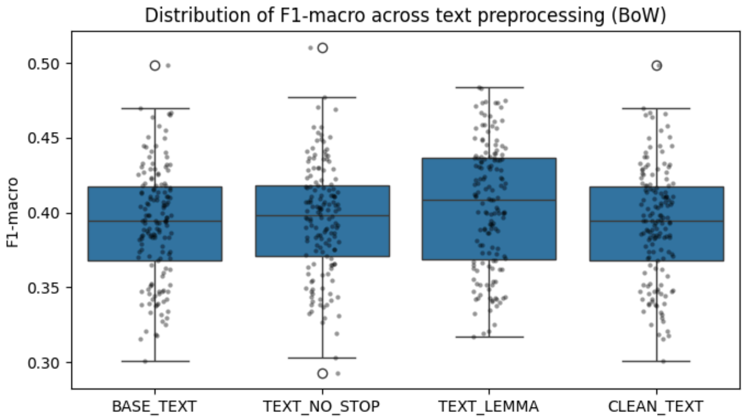
II. Text Preprocessing

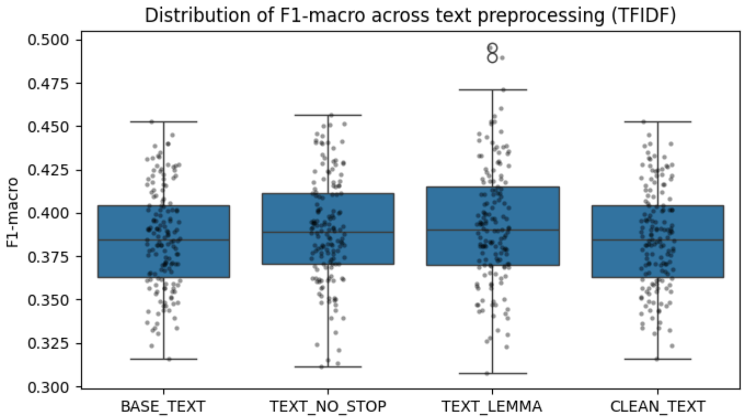
Bow:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	LogisticRegression	0.5649 ± 0.021	0.4592 ± 0.0498	0.4164 ± 0.0249	0.4259 ± 0.03
NO_STOP	LogisticRegression	0.5661 ± 0.0186	0.4593 ± 0.0464	0.4163 ± 0.0216	0.426 ± 0.0264
LEMMA	LogisticRegression	**0.582 ± 0.0218**	**0.487 ± 0.0462**	**0.4318 ± 0.0202**	**0.4435 ± 0.0231**
CLEAN	LogisticRegression	0.5649 ± 0.021	0.4592 ± 0.0498	0.4164 ± 0.0249	0.4259 ± 0.03

TFIDF:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	XGBClassifier	0.5604 ± 0.0216	0.4586 ± 0.0365	0.3976 ± 0.0194	0.4124 ± 0.0224
NO_STOP	XGBClassifier	0.5639 ± 0.0185	0.4691 ± 0.0528	0.4019 ± 0.0188	0.4173 ± 0.0231
LEMMA	XGBClassifier	**0.5807 ± 0.0172**	**0.5046 ± 0.0745**	**0.4097 ± 0.0248**	**0.4267 ± 0.0309**
CLEAN	XGBClassifier	0.5604 ± 0.0216	0.4586 ± 0.0365	0.3976 ± 0.0194	0.4124 ± 0.0224

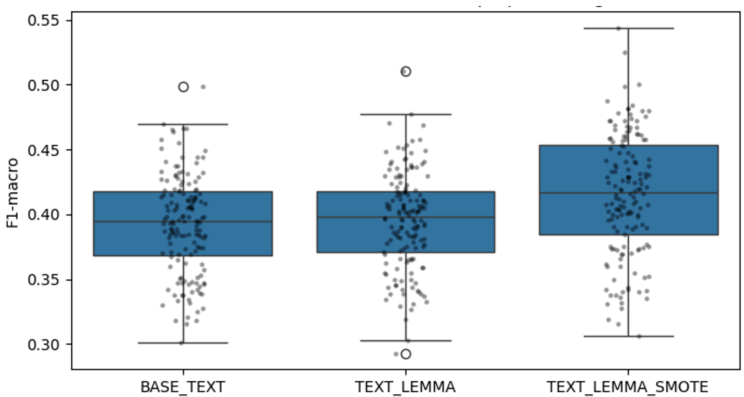




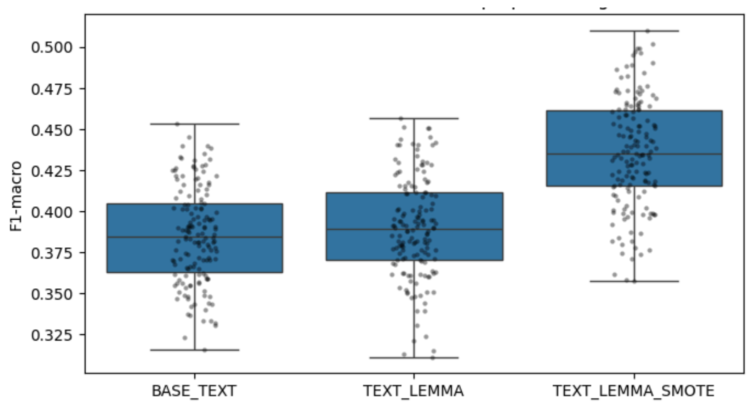
III. Balancing with SMOTE

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LogisticRegression	0.5604 ± 0.023	0.4505 ± 0.0258	0.4537 ± 0.032	0.4485 ± 0.027
TFIDF	LogisticRegression	**0.5902 ± 0.0173**	**0.475 ± 0.0316**	**0.4684 ± 0.0259**	**0.4671 ± 0.026**

BoW:

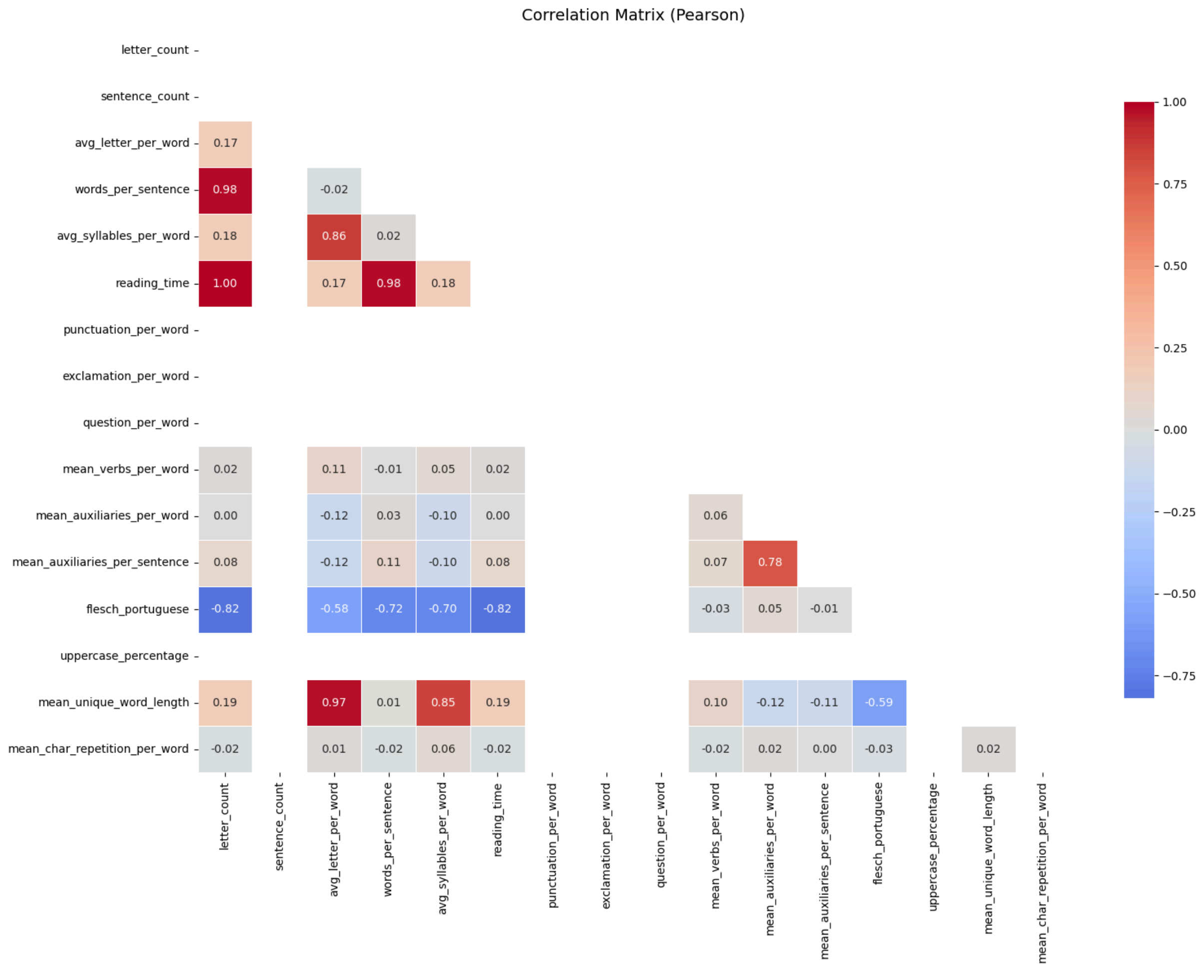


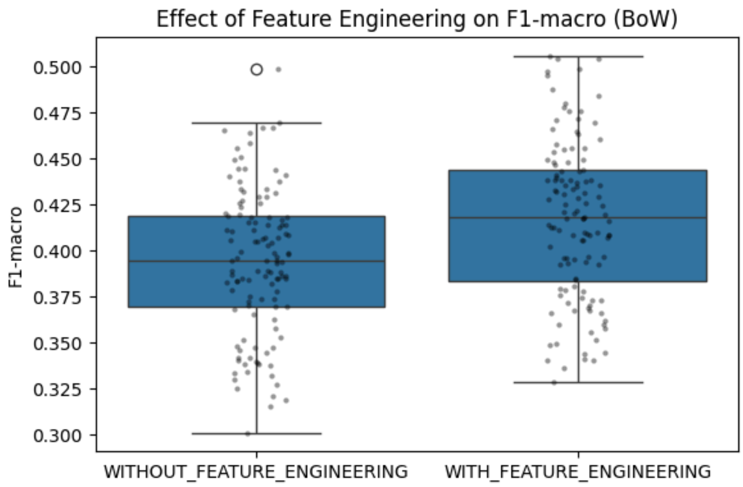
TFIDF:

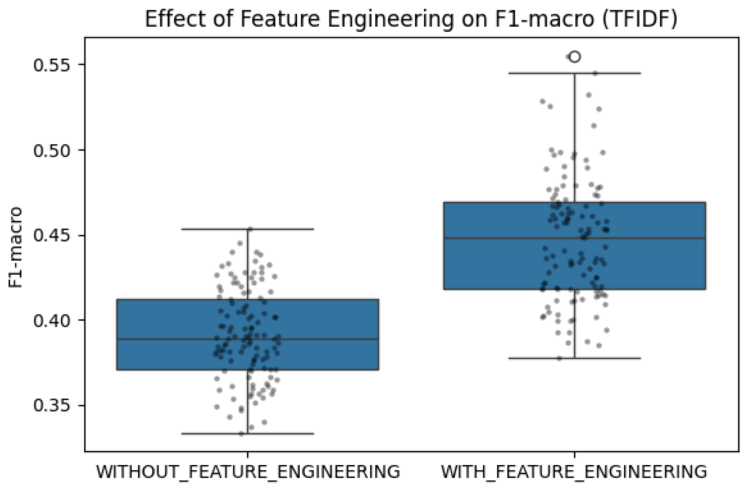


IV. Feature Engineering

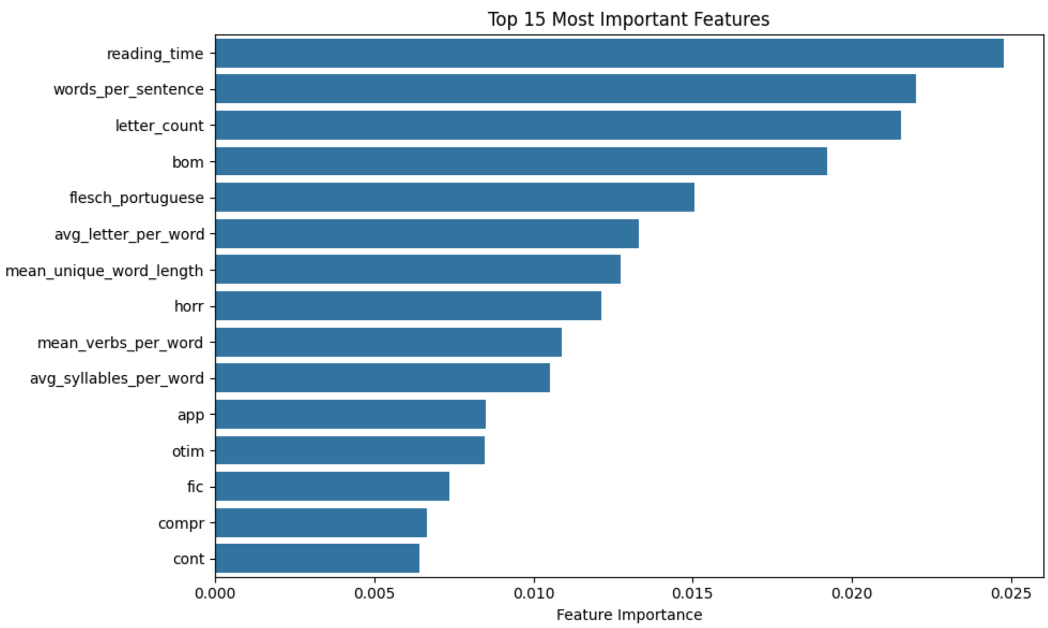
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LogisticRegression	0.5694 ± 0.02	0.4609 ± 0.0487	0.4595 ± 0.0299	0.4551 ± 0.0306
TFIDF	LogisticRegression	**0.5983 ± 0.0136**	**0.4764 ± 0.0359**	**0.4923 ± 0.0307**	**0.479 ± 0.0291**

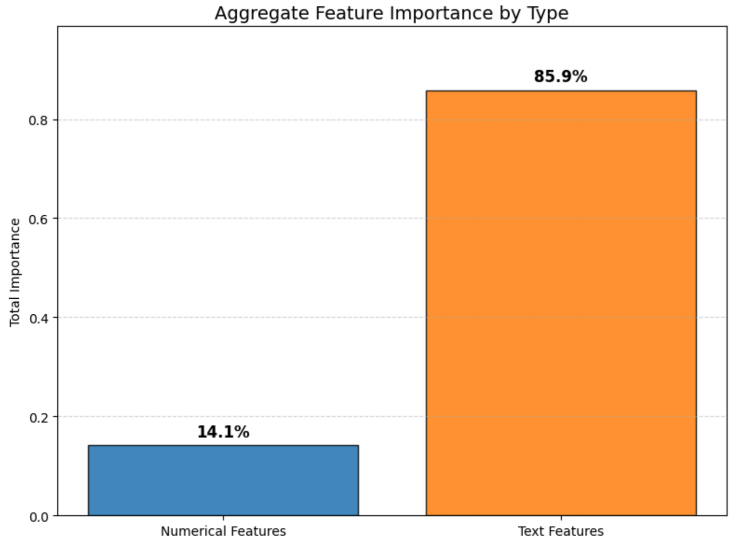



Effect of Feature Engineering on F1-macro (BoW)
F1-macro
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
WITHOUT_FEATURE_ENGINEERING
WITH_FEATURE_ENGINEERING

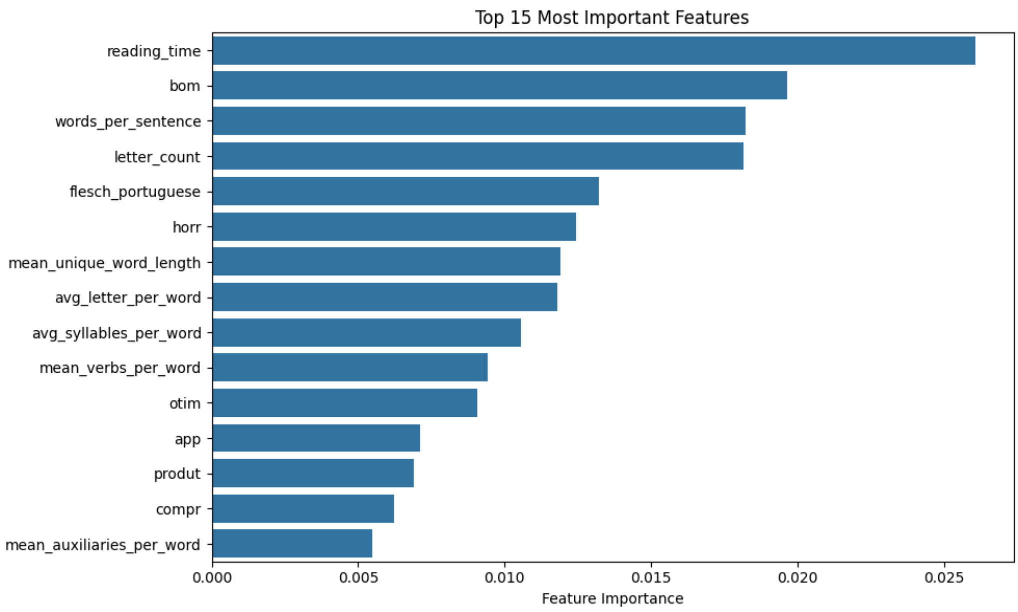

Effect of Feature Engineering on F1-macro (TFIDF)
F1-macro
0.55
0.50
0.45
0.40
0.35
WITHOUT_FEATURE_ENGINEERING
WITH_FEATURE_ENGINEERING

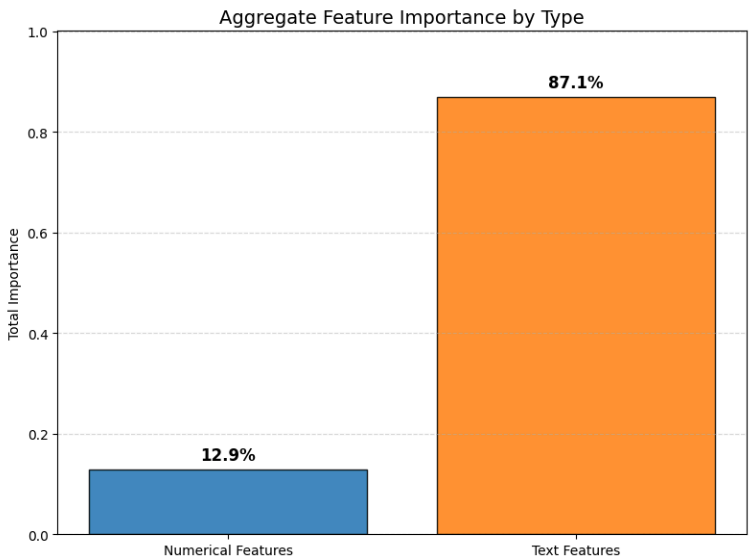
BoW:


Top 15 Most Important Features
reading_time
words_per_sentence
letter_count
bom
flesch_portuguese
avg_letter_per_word
mean_unique_word_length
horr
mean_verbs_per_word
avg_syllables_per_word
app
otim
fic
compr
cont
0.000
0.005
0.010
0.015
0.020
0.025
Feature Importance


Aggregate Feature Importance by Type
Total Importance
0.0
0.2
0.4
0.6
0.8
14.1%
85.9%
Numerical Features
Text Features

TFIDF:


Top 15 Most Important Features
reading_time
bom
words_per_sentence
letter_count
flesch_portuguese
horr
mean_unique_word_length
avg_letter_per_word
avg_syllables_per_word
mean_verbs_per_word
otim
app
produt
compr
mean_auxiliaries_per_word
0.000
0.005
0.010
0.015
0.020
0.025
Feature Importance


Aggregate Feature Importance by Type
Total Importance
0.0
0.2
0.4
0.6
0.8
1.0
12.9%
87.1%
Numerical Features
Text Features

V. Final Test

Argument	**Value**
Model	LogisticRegression
Representation	TF-IDF
Stopwords or Lemmatization?	Lemmatization
Additional Features?	Sim
F1 macro (CV)	0.479 ± 0.0291

Accuracy	**Precision_macro**	**Recall_macro**	**F1_macro**
0.5965	0.451	0.457	0.4532

VI. BERTimbau Evaluation

Without balancing:

Loss	**Accuracy**	**Precision_macro**	**Recall_macro**	**F1_macro**
0.9586	**0.6608**	**0.452**	0.4579	0.4532

With balancing (CB Loss):

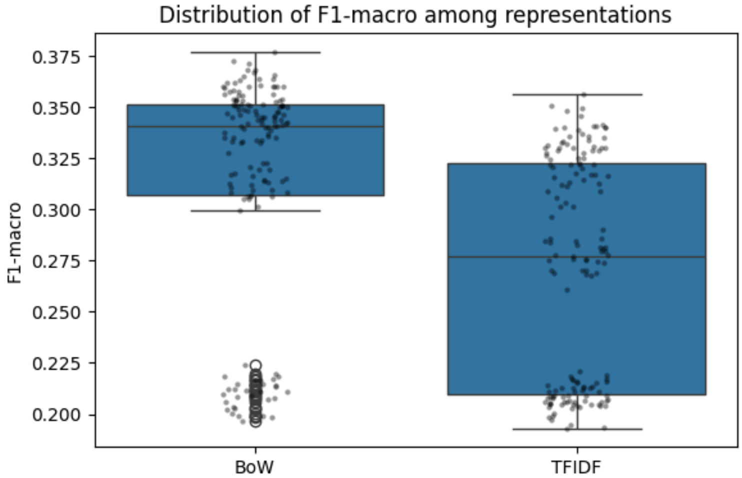
Loss	**Accuracy**	**Precision_macro**	**Recall_macro**	**F1_macro**
1.1728	0.5831	0.4417	**0.479**	**0.4539**

6. GoEmotions

I. Baseline (with BASE_TEXT)

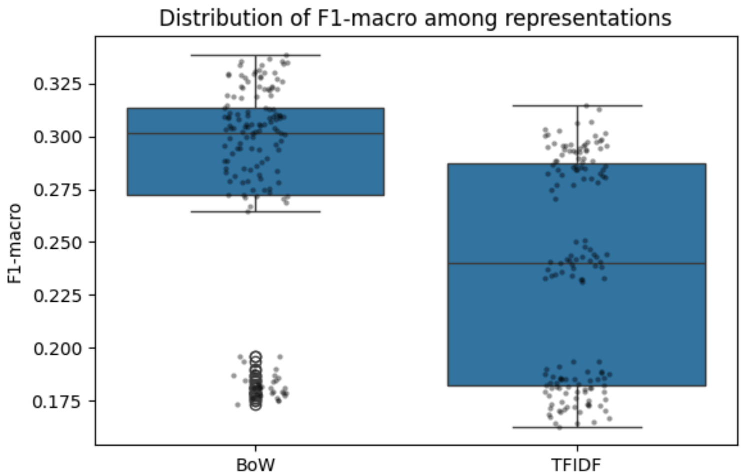
English:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LinearSVC	**0.3061 ± 0.0047**	0.4379 ± 0.0118	**0.3145 ± 0.0073**	**0.359 ± 0.0076**
TFIDF	XGBClassifier	0.2736 ± 0.0255	**0.594 ± 0.0171**	0.2646 ± 0.0074	0.3361 ± 0.0086



Portuguese:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LinearSVC	**0.2919 ± 0.0039**	0.4142 ± 0.0137	**0.2833 ± 0.0054**	0.3283 ± 0.0059
TFIDF	XGBClassifier	0.2264 ± 0.0059	0.5808 ± 0.0187	0.2265 ± 0.0052	0.2969 ± 0.0073



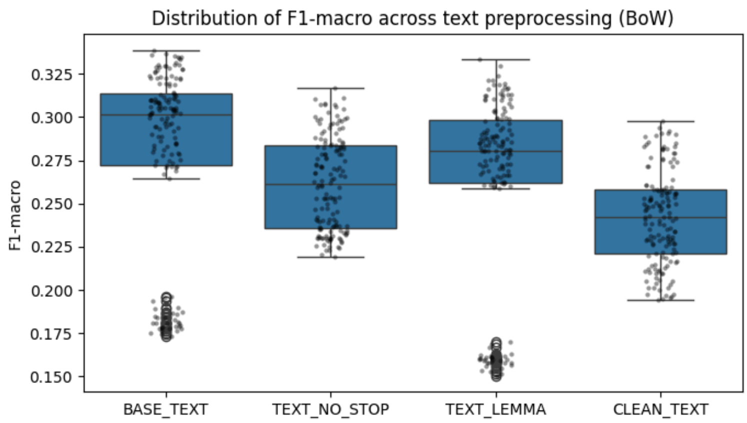
II. Text Preprocessing

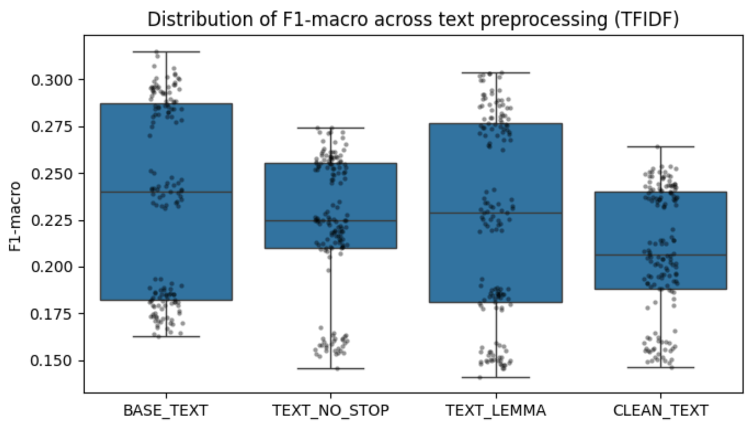
Bow:

	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	LinearSVC	**0.2919 ± 0.0039**	**0.4142 ± 0.0137**	**0.2833 ± 0.0054**	**0.3283 ± 0.0059**
NO_STOP	LinearSVC	0.237 ± 0.0039	0.3931 ± 0.0157	0.251 ± 0.0072	0.2989 ± 0.0084
LEMMA	LinearSVC	0.2853 ± 0.006	0.392 ± 0.0139	0.2709 ± 0.0063	0.3136 ± 0.0075
CLEAN	LinearSVC	0.2262 ± 0.0047	0.3726 ± 0.012	0.2357 ± 0.0071	0.2818 ± 0.0077

TFIDF:

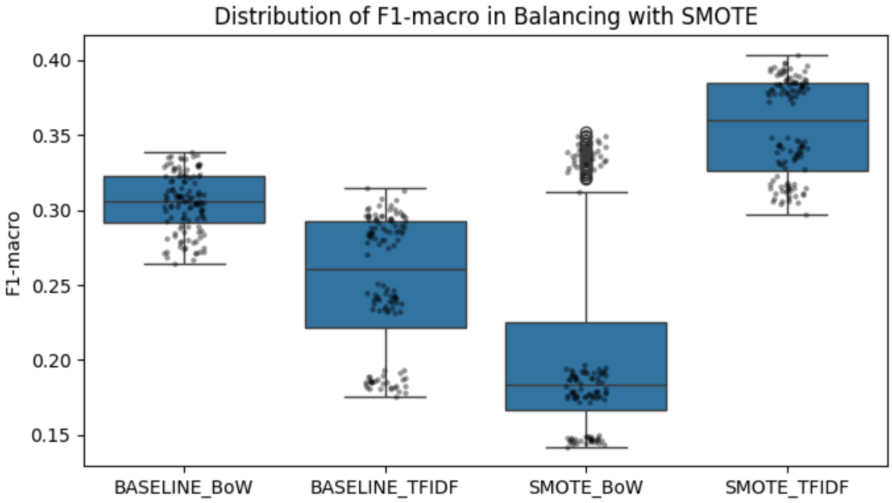
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BASE	XGBClassifier	**0.2264 ± 0.0059**	**0.5808 ± 0.0187**	**0.2265 ± 0.0052**	**0.2969 ± 0.0073**
NO_STOP	XGBClassifier	0.1496 ± 0.0044	0.5747 ± 0.0169	0.1911 ± 0.005	0.2602 ± 0.007
LEMMA	XGBClassifier	0.2415 ± 0.0164	0.5656 ± 0.0258	0.2221 ± 0.0073	0.2897 ± 0.0095
CLEAN	XGBClassifier	0.1504 ± 0.0056	0.5495 ± 0.0218	0.18 ± 0.0053	0.2441 ± 0.0072





III. Balancing with SMOTE[5]

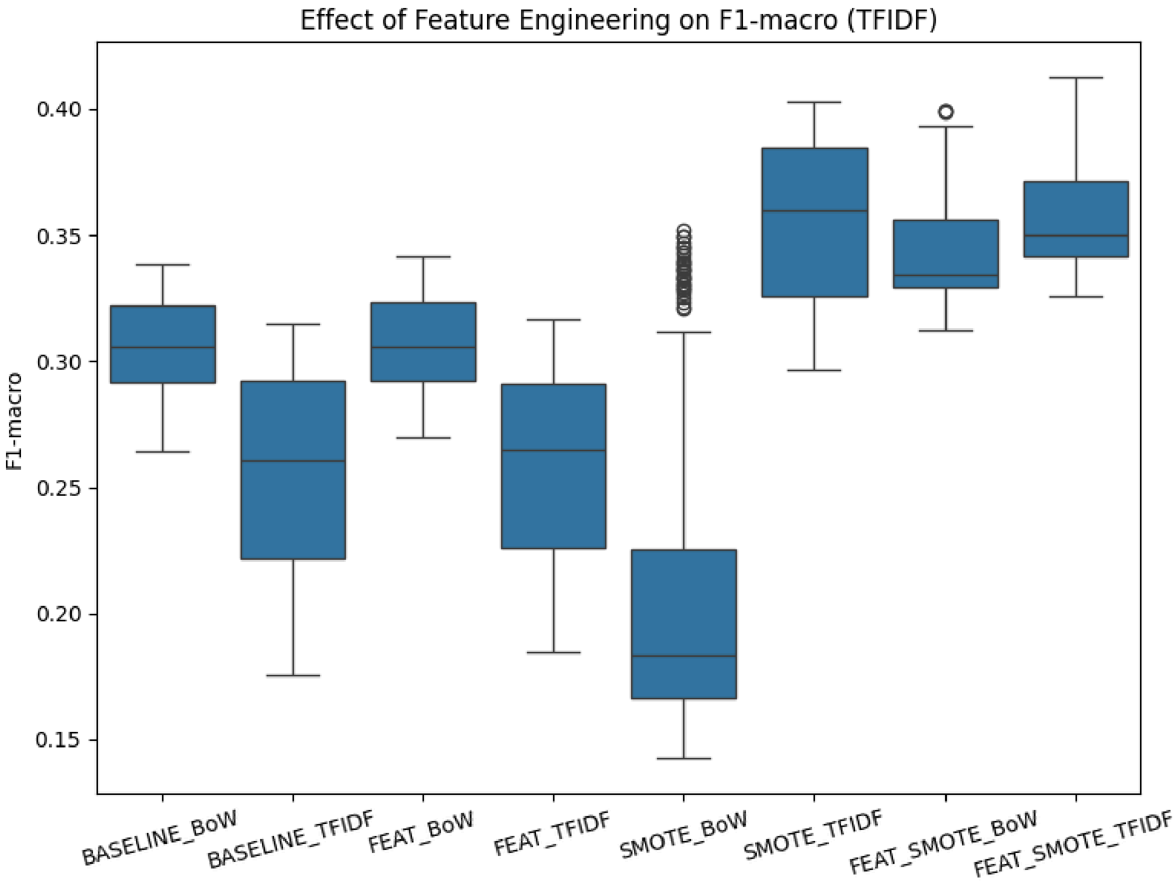
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	XGBClassifier	0.3069 ± 0.0062	**0.5387 ± 0.0226**	0.2784 ± 0.0079	0.3349 ± 0.0094
TFIDF	XGBClassifier	**0.3163 ± 0.0054**	0.4163 ± 0.0104	**0.3787 ± 0.0081**	0.3874 ± 0.0072

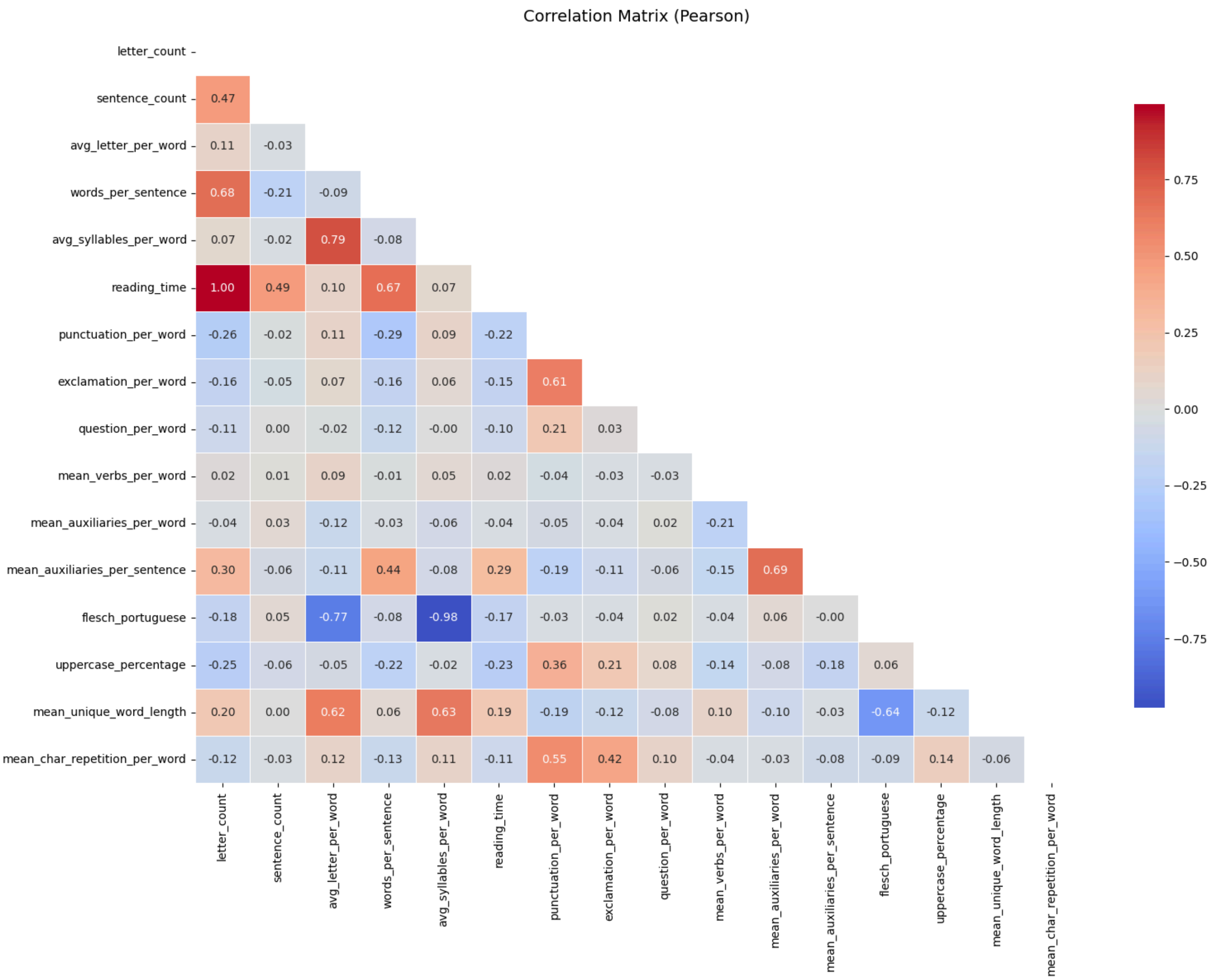


[5] Without Random Forest.

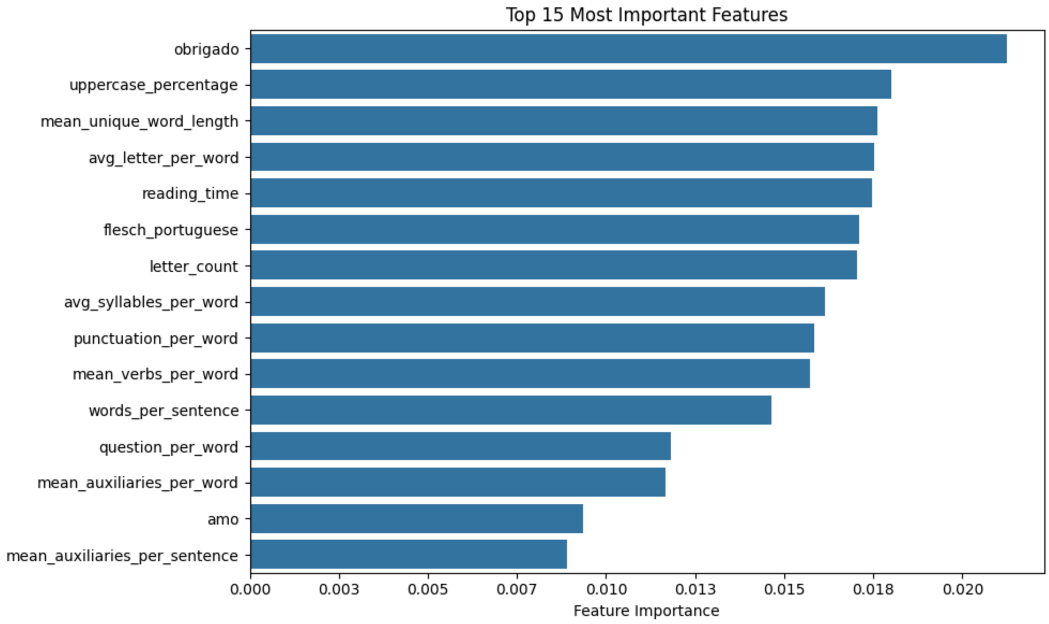
IV. Feature Engineering

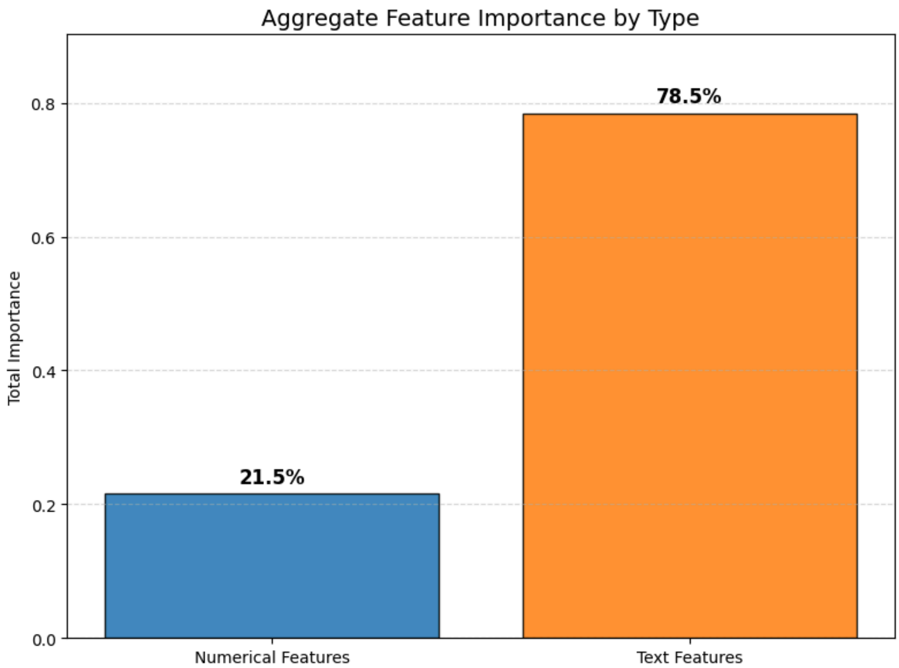
	Model	Accuracy	Precision_macro	Recall_macro	F1_macro
BoW	LinearSVC	**0.2952 ± 0.0039**	0.4233 ± 0.0133	0.2819 ± 0.0051	0.3302 ± 0.0058
TFIDF	XGBClassifier	0.2562 ± 0.0065	**0.5603 ± 0.0297**	0.2262 ± 0.0075	0.2961 ± 0.0101
BoW SMOTE	LogisticRegression	0.2324 ± 0.004	0.3296 ± 0.0065	0.4819 ± 0.0074	0.3871 ± 0.0059
TFIDF SMOTE	LogisticRegression	0.2065 ± 0.0039	0.326 ± 0.0064	**0.5431 ± 0.008**	**0.3996 ± 0.0064**



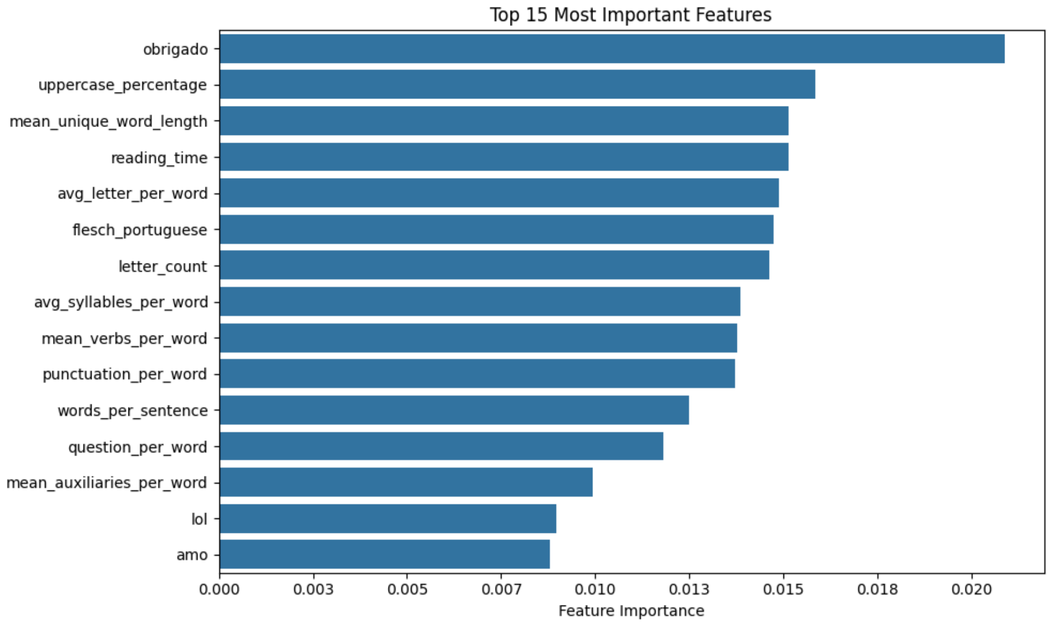

Correlation Matrix (Pearson)
letter_count
sentence_count 0.47
avg_letter_per_word 0.11 -0.03
words_per_sentence 0.68 -0.21 -0.09
avg_syllables_per_word 0.07 -0.02 0.79 -0.08
reading_time 1.00 0.49 0.10 0.67 0.07
punctuation_per_word -0.26 -0.02 0.11 -0.29 0.09 -0.22
exclamation_per_word -0.16 -0.05 0.07 -0.16 0.06 -0.15 0.61
question_per_word -0.11 0.00 -0.02 -0.12 -0.00 -0.10 0.21 0.03
mean_verbs_per_word 0.02 0.01 0.09 -0.01 0.05 0.02 -0.04 -0.03 -0.03
mean_auxiliaries_per_word -0.04 0.03 -0.12 -0.03 -0.06 -0.04 -0.05 -0.04 0.02 -0.21
mean_auxiliaries_per_sentence 0.30 -0.06 -0.11 0.44 -0.08 0.29 -0.19 -0.11 -0.06 -0.15 0.69
flesch_portuguese -0.18 0.05 -0.77 -0.08 -0.98 -0.17 -0.03 -0.04 0.02 -0.04 0.06 -0.00
uppercase_percentage -0.25 -0.06 -0.05 -0.22 -0.02 -0.23 0.36 0.21 0.08 -0.14 -0.08 -0.18 0.06
mean_unique_word_length 0.20 0.00 0.62 0.06 0.63 0.19 -0.19 -0.12 -0.08 0.10 -0.10 -0.03 -0.64 -0.12
mean_char_repetition_per_word -0.12 -0.03 0.12 -0.13 0.11 -0.11 0.55 0.42 0.10 -0.04 -0.03 -0.08 -0.09 0.14 -0.06
letter_count
sentence_count
avg_letter_per_word
words_per_sentence
avg_syllables_per_word
reading_time
punctuation_per_word
exclamation_per_word
question_per_word
mean_verbs_per_word
mean_auxiliaries_per_word
mean_auxiliaries_per_sentence
flesch_portuguese
uppercase_percentage
mean_unique_word_length
mean_char_repetition_per_word
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

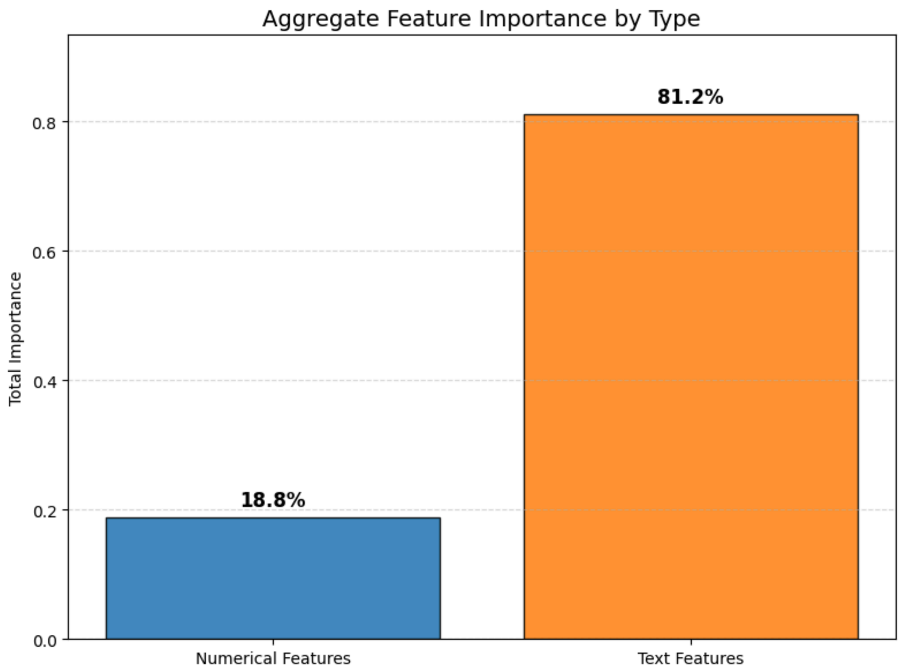
BoW:


Top 15 Most Important Features
obrigado
uppercase_percentage
mean_unique_word_length
avg_letter_per_word
reading_time
flesch_portuguese
letter_count
avg_syllables_per_word
punctuation_per_word
mean_verbs_per_word
words_per_sentence
question_per_word
mean_auxiliaries_per_word
amo
mean_auxiliaries_per_sentence
0.000
0.003
0.005
0.007
0.010
0.013
0.015
0.018
0.020
Feature Importance


Aggregate Feature Importance by Type
Total Importance
0.0
0.2
0.4
0.6
0.8
21.5%
78.5%
Numerical Features
Text Features

TFIDF:

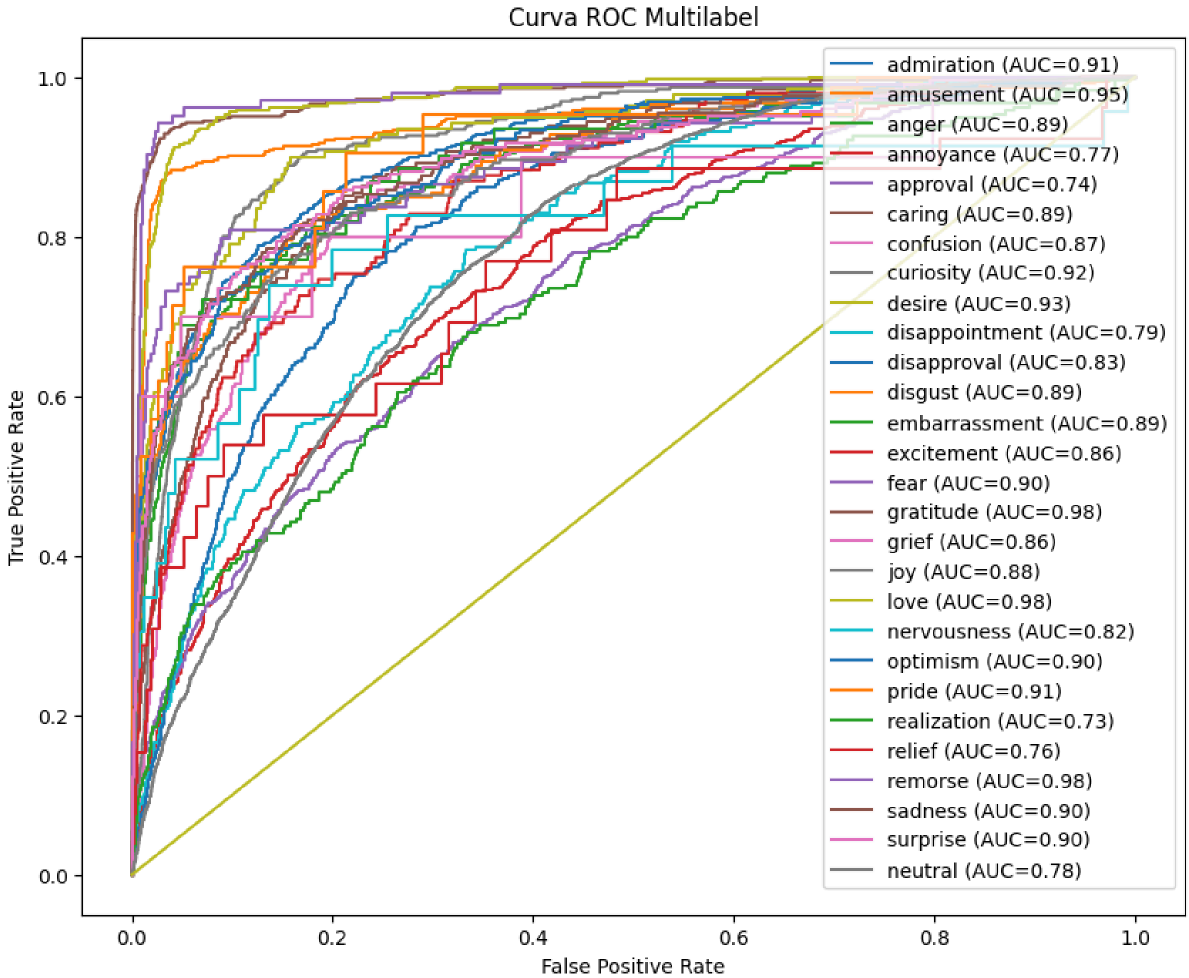

Top 15 Most Important Features
obrigado
uppercase_percentage
mean_unique_word_length
reading_time
avg_letter_per_word
flesch_portuguese
letter_count
avg_syllables_per_word
mean_verbs_per_word
punctuation_per_word
words_per_sentence
question_per_word
mean_auxiliaries_per_word
lol
amo
0.000
0.003
0.005
0.007
0.010
0.013
0.015
0.018
0.020
Feature Importance


Aggregate Feature Importance by Type
Total Importance
0.0
0.2
0.4
0.6
0.8
18.8%
81.2%
Numerical Features
Text Features

V. Final Test

Argument	Value
Model	LogisticRegression
Representation	TF-IDF
Stopwords or Lemmatization?	Não
Additional Features?	Sim
F1 macro (CV)	0.6822 ± 0.0925

Accuracy	Precision_macro	Recall_macro	F1_macro
0.1874	0.3096	0.5527	0.3881



VI. BERTimbau Evaluation

Without balancing:

Loss	Accuracy	Precision_macro	Recall_macro	F1_macro
0.0966	**0.4201**	0.5087	0.3152	0.3583

With balancing (CB Loss):

Loss	Accuracy	Precision_macro	Recall_macro	F1_macro
0.0644	0.3929	**0.5430**	**0.3548**	**0.4013**

****